

23AHS21 - probability & statistics
Unit - IV : Estimation and Testing of Hypothesis, Large Sample Tests

Assignment - IV

- 1) Define
i) Estimation and its types.
ii) Null hypothesis, alternate hypothesis.
iii) Critical region, acceptance region.
iv) Level of significance, maximum error of estimate.
v) Type-I error, Type-II error, one-tailed test, Two-tailed test. [C04 : Understand]
- 2) What is the size of the smallest sample required to estimate the an unknown proportion to within a maximum error of 0.06 with atleast 95% Confidence. [C04 : Apply]
- 3) The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D 2.5 inches. [C04 : Apply]
- 4) Random samples of 400 men and 600 women were asked whether they would like to have flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level. [C04 : Apply]

- 5) i) Among 900 people in a state 90 are found to be chapati-eaters. Construct 99% Confidence interval for the true proportion.
- ii) The mean and the standard deviation of a population are 11795 and 14054 respectively. If $n=50$, find 95% Confidence interval for the mean.
- [C04: Apply]